

CHALMERS



Finite Element Method - Structures, TME245

Computer exercise 1: 2D elastic analysis with linear triangular elements

Instructions

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Preface

Within this computer exercise, a 2D elasticity problem is considered. In the oral report, you should be able to explain any theory asked for, and also explain the MATLAB/Python implementations made. No report is to be submitted.

To generate the discretised geometry of the problem, the function `TunnelMeshGen` is provided in both MATLAB and Python format. Please check the documentation at the beginning of each file for a description of the inputs and outputs.

Learning outcomes: The learning outcomes of this project are to study the FE-implementation of an elasticity problem in 2D, and to implement a three-noded constant strain element, and study convergence of FE-solutions.

It will be possible to do the oral report already at the computer class on Friday 24 January. To do so, please simply write your names on the help lists in the computer rooms.

For the final day of reporting (26 January), we have scheduled reporting time slots, to which you can sign up here:

<https://chooodle.portal.chalmers.se/JGKEpyIvAlTuaSjo>

Please make sure to sign-up using your group number! Please note that each group is only allowed to reserve one time slot, and that each time slot is maximised to 13 groups.

Göteborg in January 2026

Martin Fagerström

Problem definition

We shall in this project study the stress and displacement field around a concrete tunnel submerged in a river at the distance Z (to the top of the tunnel) from the average water surface, cf. Figure 1.

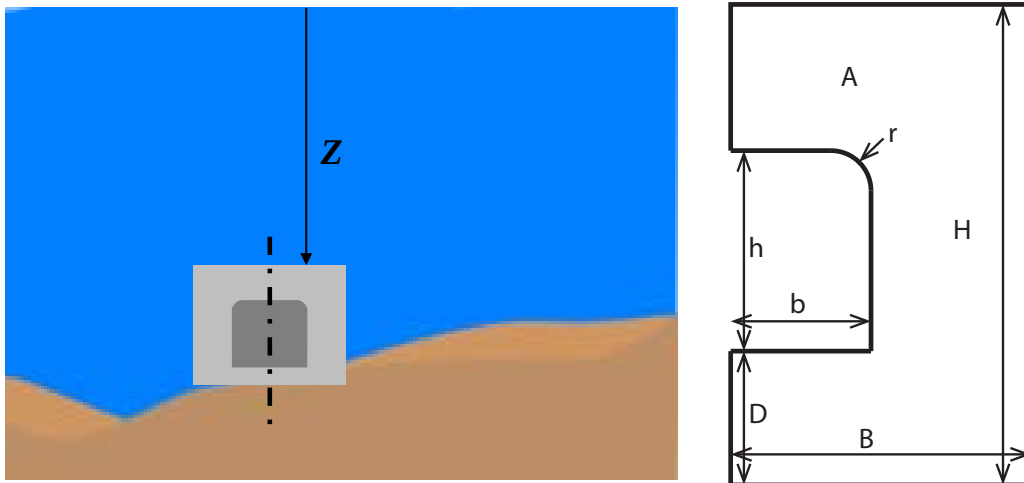


Figure 1: a) A tunnel submerged in water and b) a close-up of half of the tunnel cross section

- The concrete material is assumed to be linearly elastic with the material properties $E = 30 \text{ GPa}$ and $\nu = 0.2$.
- The volume load intensity of the tunnel is $\mathbf{b} = [0, -\rho g]^T \text{ N/m}^3$ directed in the negative y -direction. The concrete density is $\rho = 2\,200 \text{ kg/m}^3$.
- The water pressure is to be approximated as constant over all outer surfaces of the tunnel, and is thereby taken as $p = Z\rho g \text{ N/m}^2$, with Z according to the table below. Please choose an appropriate water density.
- The geometry under consideration is defined by the geometry parameters as defined in the figure. For clarity, we will be considering a section of the tunnel somewhere along its length, which means that without loss of generality the out-of-plane thickness of the 2D problem can be set to $L = 1 \text{ m}$. Further numerical values for geometry parameters are given in the table below:

Z	50 m
B	15 m
H	22 m
D	8 m
b	6 m
h	7 m
r	2 m

Task 1: Elastic analysis in 2D using MATLAB/Python and CALFEM

Complete the following tasks and comment on the results when appropriate.

- Determine and state appropriate boundary conditions** for the half-domain A shown to the right in Figure 1. Please consider the tunnel to be firmly attached to the ground. **Give the strong form of the problem at hand**, in particular, choose whether the situation at hand is relevant to the **plane strain or the plane stress assumption**.
- Given the FE-form of the 2D linear elastic problem (not necessary to show or derive), **implement your own function file for a three noded so-called constant strain element** that calculates the element stiffness matrix and the element volume load vector (the latter due to gravity load). Please check the lecture notes for coding suggestions.

To verify your code, an element with nodal coordinates (all in meters):

$$(x_1, y_1) = (0, 0), (x_2, y_2) = (1, 0.25), (x_3, y_3) = (0.5, 1)$$

yields, for the loading conditions and material parameters above, the following element stiffness matrix and element load vector:

$$K_e = \begin{bmatrix} 1.2500 & 0.4464 & -1.2500 & -0.1786 & -0.0000 & -0.2679 \\ 0.4464 & 0.8780 & 0.0298 & -0.0595 & -0.4762 & -0.8185 \\ -1.2500 & 0.0298 & 2.0833 & -0.5952 & -0.8333 & 0.5655 \\ -0.1786 & -0.0595 & -0.5952 & 1.1905 & 0.7738 & -1.1310 \\ -0.0000 & -0.4762 & -0.8333 & 0.7738 & 0.8333 & -0.2976 \\ -0.2679 & -0.8185 & 0.5655 & -1.1310 & -0.2976 & 1.9494 \end{bmatrix} \cdot 10^{10} \text{ N/m}$$

$$f_e = \begin{bmatrix} 0 \\ -3.1474 \\ 0 \\ -3.1474 \\ 0 \\ -3.1474 \end{bmatrix} \cdot 10^3 \text{ N}$$

- c) **Implement your own function file that computes the nodal load contributions resulting from a pressure load applied along an external element edge.** Sufficient arguments for the function to compute the resulting nodal load contributions should be the coordinates of the nodes sharing an edge segment and the prescribed traction given by the pressure ($\underline{t} = [t_x, t_y]^T$).

For the use in task d), please note that `RightSide_nodes` lists the nodes in order from bottom to top, and that `TopSide_nodes` lists the nodes in order from right to left.

To verify your implementation, you can compare the results to the case with an edge segment of unit thickness $h = 1$ m with (local) coordinates (also in meters) $(x_1, y_1) = (0, 0)$ and $(x_2, y_2) = (1, 1)$, and an applied pressure $p = 5$ Pa (See Figure 2 below). This results in the following nodal load contributions:

$$f_{1,x} = -2.5 \text{ N}, f_{1,y} = 2.5 \text{ N}, f_{2,x} = -2.5 \text{ N}, f_{2,y} = 2.5 \text{ N}.$$

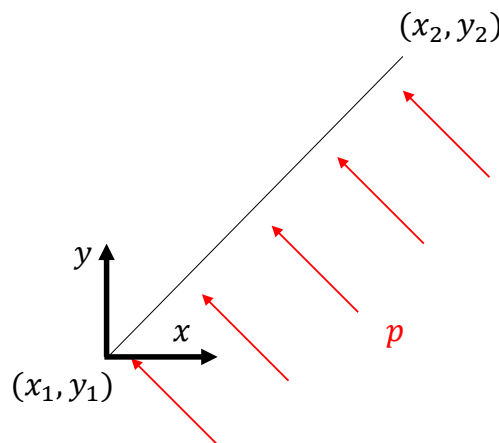


Figure 2: Example segment for code verification of pressure loads

To further verify your computation of the total pressure load contribution, compare the total force components (in x - and y -direction) resulting from the pressure on the entire exterior surface (computed by pen and paper) to the sum of all nodal forces in the x - and y -directions, respectively.

- d) Solve the elasticity problem considering gravity and the effect of the constant water pressure. Use the function `TunnelMeshGen` to **generate a mesh consisting of linear triangular elements (CST)**. Please note that since there is no `Dof`-matrix as output you may need to construct that

on your own given that the numbering of the degrees of freedom (in $\mathbf{a}$) are related to numbering of the nodes such that

$$\text{Node: } k \Rightarrow u_{x,k} = a_{2k-1}, u_{y,k} = a_{2k}$$

To visualise the discretised geometry in MATLAB, you may use the CALFEM function `eldraw2`. In Python, it is recommended to use a `PolyCollection`, please see `polygon_plot.py` for an example.

Then **write a 2D elasticity finite element program and determine the nodal displacements** using the functions written in Tasks b) and c). Please note that for this task, you may NOT use any CALFEM functions to solve for the displacements. Specifically, **compute the maximum vertical displacement of any point in the tunnel section.**

To visualise the deformed shape, **plot the deformed mesh for a discretisation of your choice.** Here, in MATLAB you may use the CALFEM function `eldisp2`. In Python, it is again recommended that you use `PolyCollection` or something similar.

Please note that the lecture notes contain several code snippets that could be very helpful in solving this task.

- e) **Implement your own function that calculates the stress components in an element midpoint.** Then implement this function in your code such that the stress state in each element midpoint is computed.

Then, based on these results, **conduct an eigenvalue analysis of the element stress state, and plot (element-wise) the maximum AND minimum principal stress distribution over the half domain.** Here, the MATLAB solution may benefit from using the function `fill`. For Python, it is recommended to once again use `PolyCollection`.

To verify your code, consider the same element example as in subtask b) above. With the nodal displacements

$$\mathbf{a}^e = \begin{bmatrix} u_{x,1} \\ u_{y,1} \\ u_{x,2} \\ u_{y,2} \\ u_{x,3} \\ u_{y,3} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0.003 \\ 0.001 \\ 0.002 \\ 0.002 \end{bmatrix}$$

the stresses becomes approximately:

$$\sigma_{xx} = 110 \text{ MPa}, \sigma_{yy} = 81 \text{ MPa}, \sigma_{zz} = 38 \text{ MPa}, \tau_{xy} = 14 \text{ MPa},$$

resulting in the principal stresses

$$\sigma_1 = 115 \text{ MPa}, \sigma_2 = 75 \text{ MPa}, \sigma_3 = 38 \text{ MPa}.$$

- f) By analysing the results, **estimate the safety factor against failure of the concrete.**

Please try to find strength values from reliable sources. Make sure to justify both your predicted value of the safety factor against failure and the source of the data when presenting your assignment.